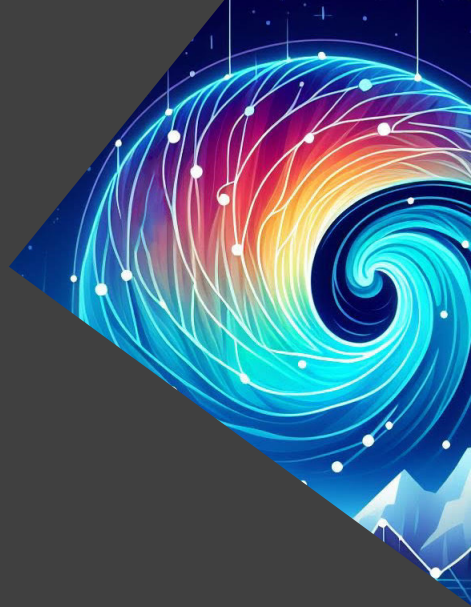


**The
Alan Turing
Institute**

**Porting the Aurora
Weather Model to Intel
Accelerated Hardware**

Iain Stenson, Rosie Wood, David
Llewellyn-Jones, Tomas Lazauskas



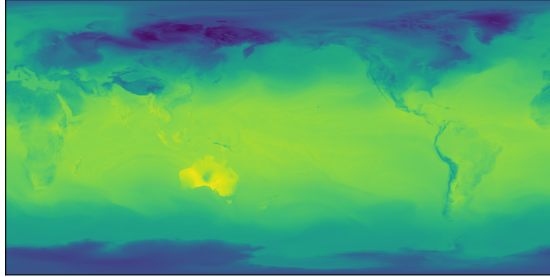
Aurora: A Foundation Model for the Earth

1. Predicts atmospheric variables
2. Adapts to specialised forecasting tasks
3. Microsoft Cambridge

1. Ease of porting
2. Similarity of results
3. Speed

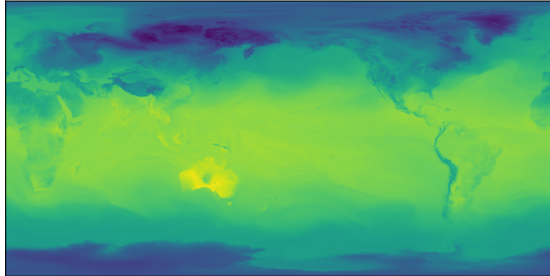
Aurora Prediction A100

2023-01-08 06:00:00

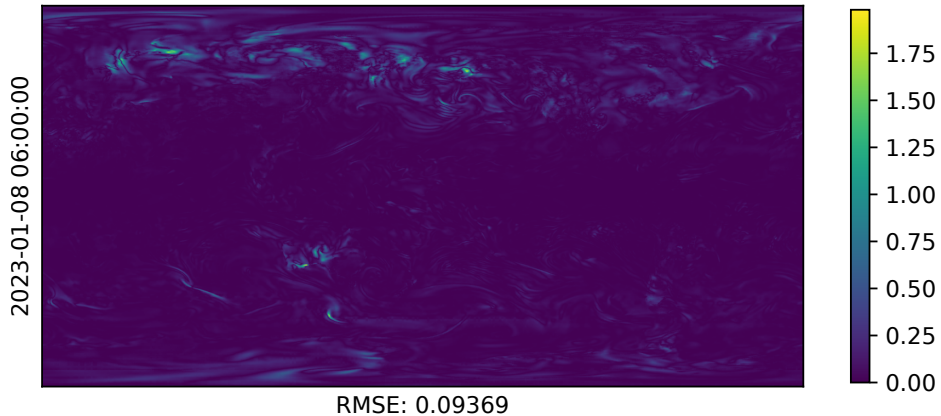


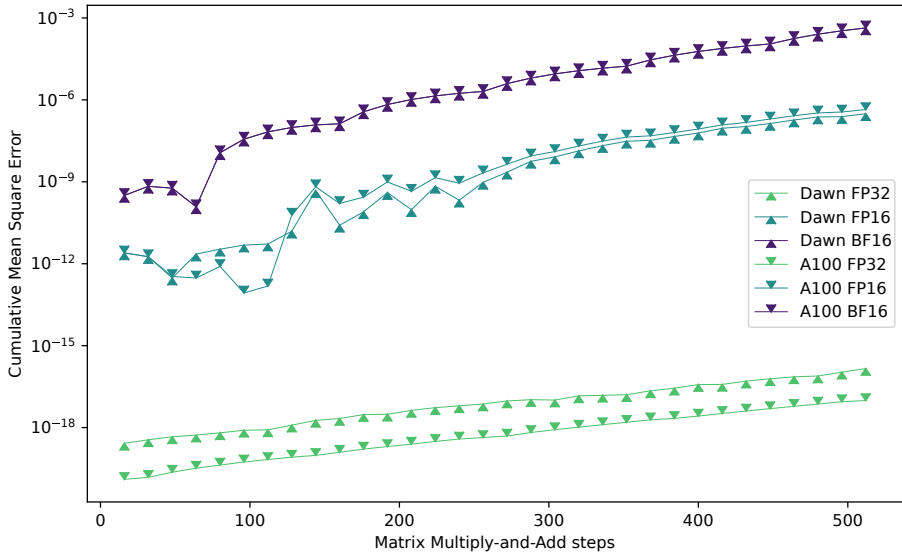
Aurora Prediction Dawn

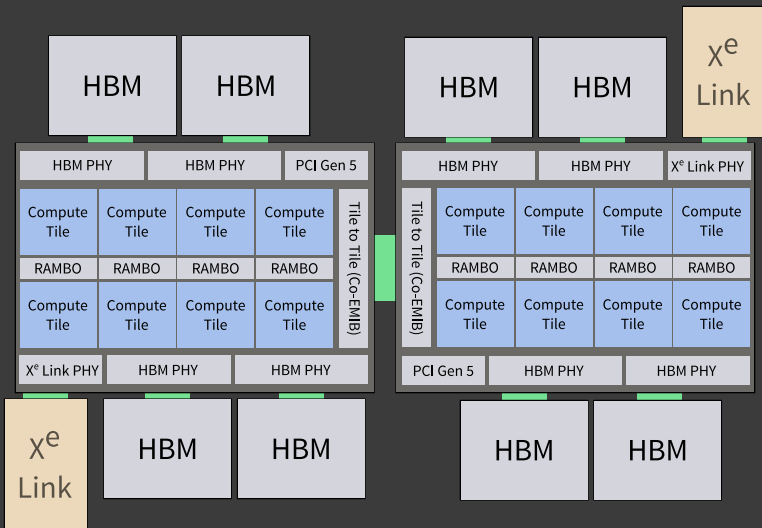
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Dawn vs. A100 on Predictions







Fine-tuning Timings

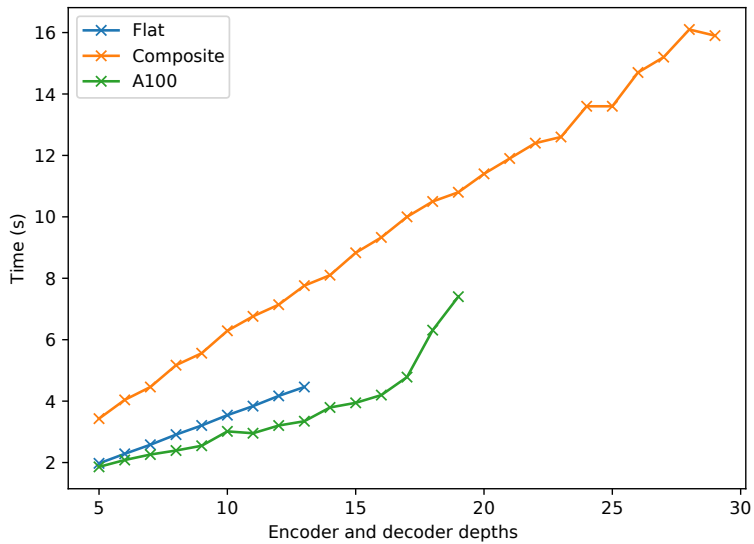
GPUs	Devices	Hierarchy	Compute time per iteration /s	Wall-clock time per iteration /s	Total run time /s
1	1	<i>flat</i>	6.86	6.86	257.43
1	2	<i>flat</i>	7.84	3.92	176.54
4	4	<i>flat</i>	8.13	2.03	122.72
4	8	<i>flat</i>	7.73	0.97	90.99
1	1	<i>composite</i>	11.10	11.10	391.70
2	2	<i>composite</i>	11.45	5.72	259.46
4	4	<i>composite</i>	11.54	2.89	152.56
1	1	A100	3.20	3.20	145.53
2	2	A100	3.25	1.62	104.85
4	4	A100	3.29	0.82	81.72

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Links

Aurora <https://github.com/microsoft/aurora/>

Report [https://github.com/alan-turing-institute/
aurora-hpc](https://github.com/alan-turing-institute/aurora-hpc)

Slides [https://github.com/alan-turing-institute/
airr-at-cam-2026-aurora](https://github.com/alan-turing-institute/airr-at-cam-2026-aurora)

With thanks to the Dawn and Baskerville teams.